Cloud Computing Serverless Functions

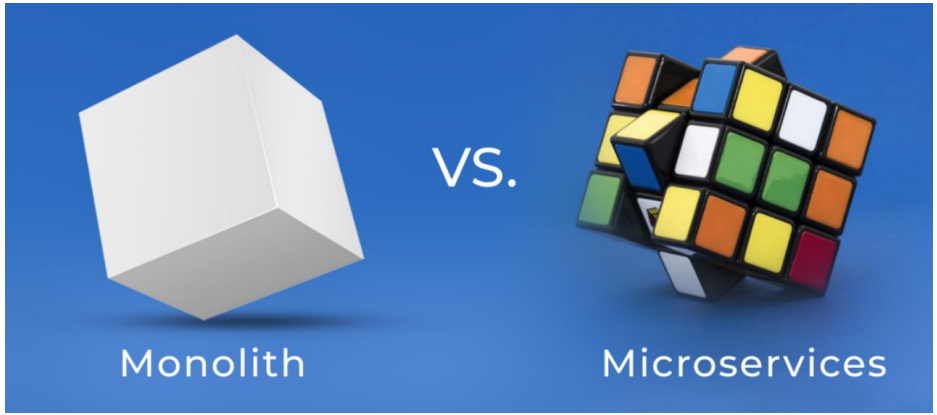
CIS437
Erik Fredericks // frederer@gvsu.edu

Adapted from Google Cloud Computing Foundations, Overview of Cloud Computing (Wufka & Canonico)

AWS CODE NINJA

Last time we made microservices

What was that again?



Last time we made microservices

Now let's make them even tinier!

Serverless functions!

- Google Cloud Functions
- AWS Lambda Functions
- Microsoft Azure Functions

(i.e., probably my favorite cloud service to play with)

Question

Can a microservice be a serverless function?

Where Cloud Functions fits within Google Cloud

Compute Engine	App Engine	Cloud Functions	Google Kubernetes Engine
IaaS	PaaS	Serverless logic	Hybrid
Virtual machines with industry-leading price/performance	A flexible, zero ops platform for building highly available apps	A lightweight fully managed serverless execution environment for building and connecting cloud services	Cluster manager and orchestration engine built on Google's container experience

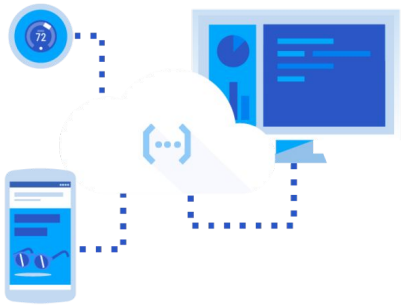
Google Cloud

Copyright Google LLC. For educational purposes in accordance with the terms of use set forth on the program Website.

The components that make Cloud Functions work

Connect and extend cloud services

Events and triggers

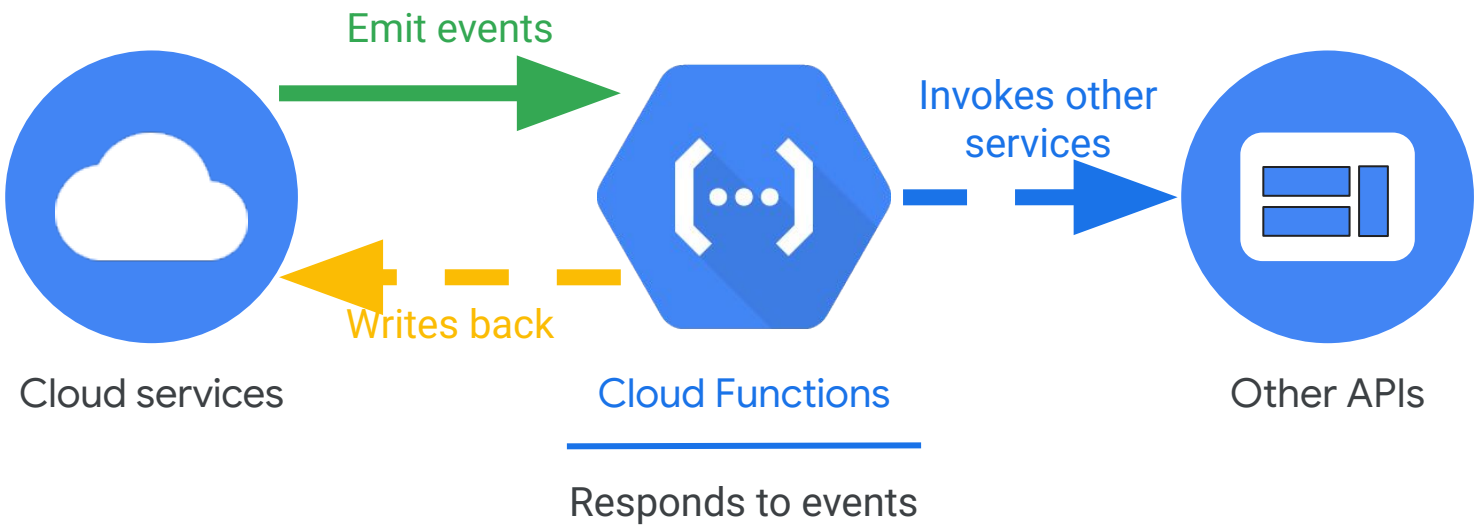


Serverless

Google Cloud

Copyright Google LLC. For educational purposes in accordance with the terms of use set forth on the program Website.

How Cloud Functions works



Google Cloud

Copyright Google LLC. For educational purposes in accordance with the terms of use set forth on the program Website.

(I was also going to include some AWS slides but)

The concepts are near-identical

¯_(ツ)_/¯

And so...

What are some applications you could use serverless functions with?

Before we dive in, considerations!

Again, with FaaS:

- No server, so no configuration
- No OS, so no setup

- Only a function, dependencies, and triggers

Practical considerations

Serverless functions typically have:

1) Limited execution time
2) Maximum amount of memory usage
3) Cost *per invocation*
4) Who is running the function?

Much like everything we've seen, these values can and will change over time

- SO IT IS UP TO YOU THE CLOUD ADMIN TO KEEP AN EYE ON THINGS

Fredericks-Si | CIS437 - Goo | CIS437 - 8 - | CIS437 - 7 - | Goog CCF - | Planck.js | Issues · notc | Content / Cl | Week 4 - De | Files | VM instance | Create functi

console.cloud.google.com/functions/add?project=cloud-apps-demos-w24

Google Cloud | cloud-apps-demos-w24 | cloud fun | Search

Cloud Functions | Create function | LEARN

URL

https://us-central1-cloud-apps-demos-w24.cloudfunctions.net/function-3

Authentication

- Allow unauthenticated invocations
  Check this if you are creating a public API or website.
- Require authentication
  Manage authorized users with Cloud IAM.

Runtime, build, connections and security settings

RUNTIME | BUILD | CONNECTIONS | SECURITY AND

Memory allocated *

- 128 MiB
- 256 MiB
- 512 MiB
- 1 GiB
- 2 GiB
- 4 GiB
- 8 GiB
- 16 GiB

CPU *
0.167

seconds

Minimum number of instances
0

Maximum number of instances
100

Runtime service account

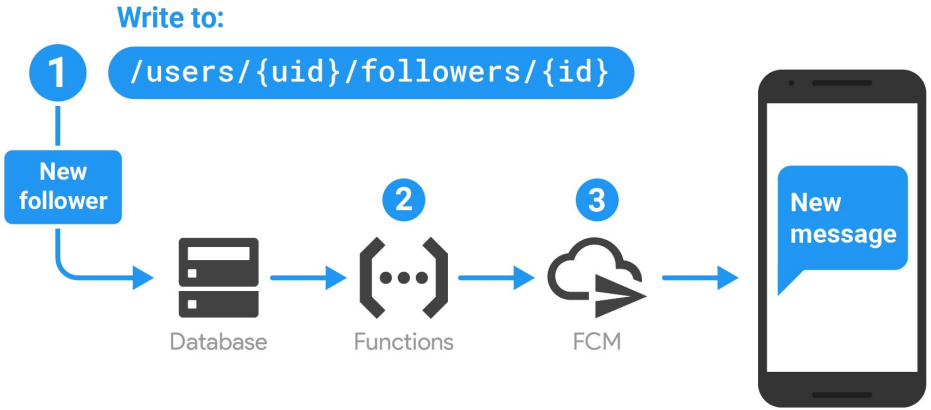
Service account *
Compute Engine default service account

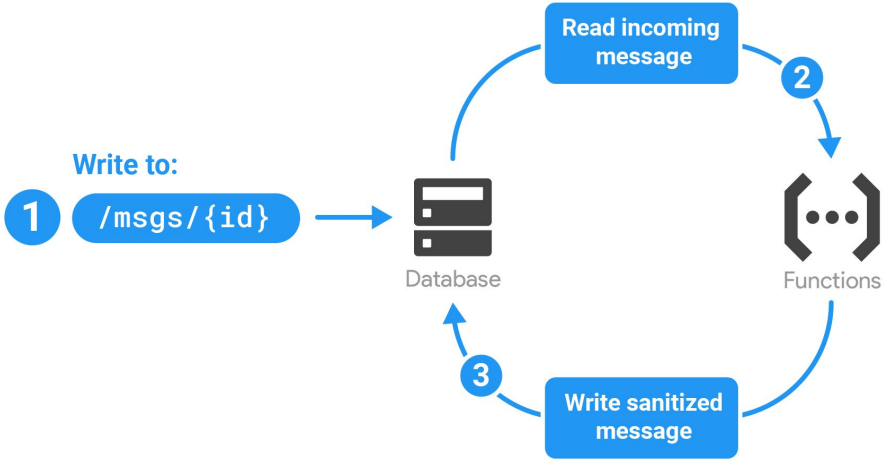
By default Cloud Functions uses the automatically created Default Compute Engine Service Account. Learn more about service accounts.

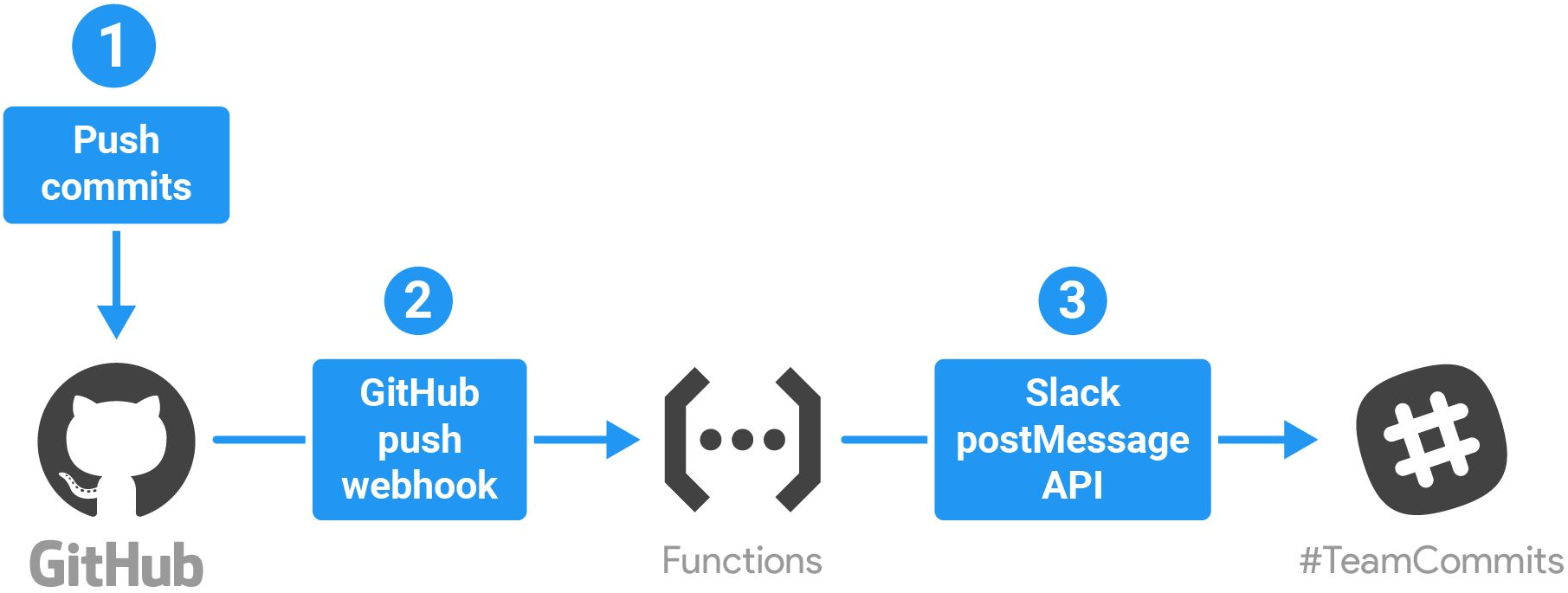
NEXT | CANCEL

https://firebase.google.com/docs/functions/use-cases

Examples!






1
Push commits
GitHub
2
GitHub push webhook
Functions
3
Slack postMessage API
#TeamCommits

Let's create one and really step through it

This is literally, going to be the most important serverless function you ever create

Shall we start with the demonstration in the terminal?

We are going to use a publicly-available API to retrieve some *mission-critical* data

In the terminal (assuming you have `curl` installed)

```
$  curl -H "Accept: application/json" https://icanhazdadjoke.com/
```

The Cloud Function

You'll notice that you have access to a lot of different languages + versions

I'm going to use Python 3.12 - you can use whatever you want

- We're going to need to replicate the `curl` command, however, so you'll need to figure that out for your particular language

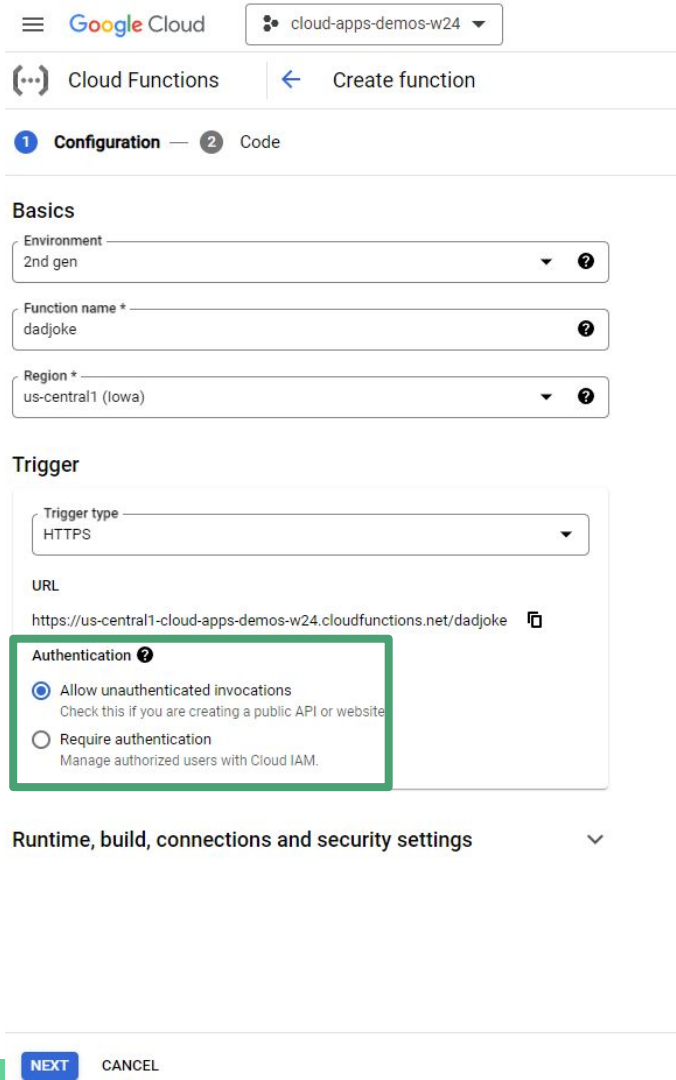
First...

Unauthenticated ➔ ANYBODY ON THE INTERNET CAN ACCESS IT!

- Pros/Cons?

Require authentication ➔ ONLY THOSE WITH ACCESS CAN ... ACCESS IT

- Pros/Cons?



jokes.py

We'll need the `requests` library

In requirements.txt, tell the CF that we need a particular version

```
requests==2.32.3
```

Currently requests>=2.32.5

jokes.py

ALWAYS USE THIS TO CHECK FOR ERRORS!!!

TEST FUNCTION

And update the default code:

```
import functions_framework
import requests

@functions_framework.http
def hello_http(request):
    url = "https://icanhazdadjoke.com"
    headers = {'Accept':'application/json'}
    r = requests.get(url, headers=headers)
    return r.json()
```

And then...

dadjoke-demo 2nd gen (Deployed at Aug 15, 2024, 4:14:49 PM) URL: https://us-central1-cloud-apps-demos-w24.cloudfunctions.net/dadjoke-demo

METRICS DETAILS SOURCE VARIABLES TRIGGER PERMISSIONS LOGS TESTING

HTTPS

URL
https://us-central1-cloud-apps-demos-w24.cloudfunctions.net/dadjoke-demo

us-central1-cloud-apps-demos-w24.cloudfunctions.net/dadjoke-demo

Pretty-print

{"id":"xHY8xsHYTnb","joke":"What lies at the bottom of the ocean and twitches? A nervous wreck.","status":200}

Let's make it purty

What's wrong?

- Just returning the entire JSON object

So...

```
joke = r.json()['joke']
return f"<h1>{joke}</h1>"
```

```
import functions_framework
import requests
import json

@functions_framework.http
def dad_joke_get(request):
    url = 'https://icanhazdadjoke.com/'
    headers = {'Accept': 'application/json'}

    response = requests.get(url=url,
headers=headers)
    response_json = response.json()

    retval = '<h1>Random Dad Joke</h1>'
    retval += f'<h2>ID: {response_json["id"]}</h2>'
    retval += f'<h2>{response_json["joke"]}</h2>'
    retval += f'<code>{response_json}</code>'
    return retval
```

Updated →

https://cloud.google.com/functions/pricing

Simple Event-Driven Function

A simple event-driven function with 128MB of memory and a 200MHz CPU, invoked 10 million times per month and running for 300ms each time using only Google APIs (no billable outbound data transfer).

Calculations

Invocations

10,000,000

Compute Time

(128 MB / 1024 MB/GB) x 0.3s = 0.0375 GB-seconds per invocation

(200 MHz / 1000 MHz/GHz) x 0.3s = 0.0600 GHz-seconds per invocation

10,000,000 invocations x 0.0375 GB-seconds = 375,000 GB-seconds per month

10,000,000 invocations x 0.0600 GHz-seconds = 600,000 GHz-seconds per month

Networking

None

Metric	Gross Value	Free Tier	Net Value	Unit Price	Total Price
Invocations	10,000,000	2,000,000	8,000,000	$0.0000004	$3.20
GB-seconds	375,000	400,000	< 0	$0.0000025	$0.00
GHz-seconds	600,000	200,000	400,000	$0.0000100	$4.00
Networking	0	5	0	$0.12	$0.00
Total / Month					$7.20

If you pay in a currency other than USD, the prices listed in your currency on Cloud Platform SKUs apply.

High Volume HTTP Function

A medium complexity HTTP Function with 256MB of memory and a 400MHz CPU, invoked 50 million times per month via HTTP, running for 500ms each time and sending 5KB of data back to the caller (5KB outbound data transfer per invocation).

Calculations

Invocations

50,000,000

Compute Time

(256 MB / 1024 MB/GB) x 0.5s = 0.125 GB-seconds per invocation

(400 MHz / 1000 MHz/GHz) x 0.5s = 0.200 GHz-seconds per invocation

50,000,000 invocations x 0.125 GB-seconds = 6,250,000 GB-seconds per month

50,000,000 invocations x 0.200 GHz-seconds = 10,000,000 GHz-seconds per month

Networking

50,000,000 invocations x (5 KB / 1024 KB/MB / 1024 MB/GB) = 238.42 GB of outbound data transfer per month

Metric	Gross Value	Free Tier	Net Value	Unit Price	Total Price
Invocations	50,000,000	2,000,000	48,000,000	$0.0000004	$19.20
GB-seconds	6,250,000	400,000	5,850,000	$0.0000025	$14.63
GHz-seconds	10,000,000	200,000	9,800,000	$0.0000100	$98.00
Networking	238.42	5	233.42	$0.12	$28.01
Total / Month					$159.84

If you pay in a currency other than USD, the prices listed in your currency on Cloud Platform SKUs apply.

FOR BALANCE

https://aws.amazon.com/lambda/pricing/

AWS Lambda Pricing

Region: US East (Ohio)

Architecture	Duration	Requests
x86 Price		
First 6 Billion GB-seconds / month	$0.0000166667 for every GB-second	$0.20 per 1M requests
Next 9 Billion GB-seconds / month	$0.000015 for every GB-second	$0.20 per 1M requests
Over 15 Billion GB-seconds / month	$0.0000133334 for every GB-second	$0.20 per 1M requests
Arm Price		
First 7.5 Billion GB-seconds / month	$0.0000133334 for every GB-second	$0.20 per 1M requests
Next 11.25 Billion GB-seconds / month	$0.0000120001 for every GB-second	$0.20 per 1M requests
Over 18.75 Billion GB-seconds / month	$0.0000106667 for every GB-second	$0.20 per 1M requests

Duration cost depends on the amount of memory you allocate to your function. You can allocate any amount of memory to your function between 128 MB and 10,240 MB, in 1 MB increments. The table below contains a few examples of the price per 1 ms associated with different memory sizes, for usage falling within the first pricing tier – for example, up to 6 billion GB-seconds / month in US East (Ohio).

x86 Price | Arm Price

Region: US East (Ohio)

Memory (MB)	Price per 1ms
128	$0.0000000021
512	$0.0000000083
1024	$0.0000000167
1536	$0.0000000250
2048	$0.0000000333
3072	$0.0000000500
4096	$0.0000000667

Interestingly, you can build them locally as well

`functions_framework`:
https://cloud.google.com/functions/docs/functions-framework

➔ https://github.com/GoogleCloudPlatform/functions-framework-python

Build your cloud functions locally, test them, and then deploy them when you're ready

...why do this?

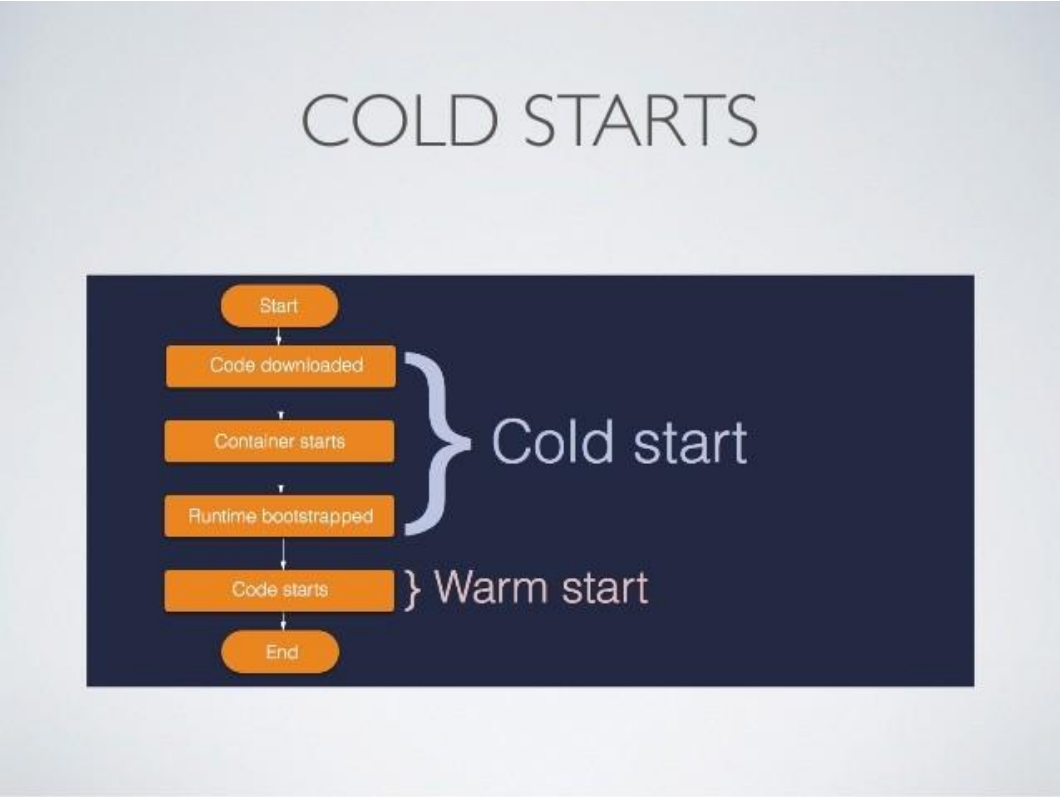
Local `functions_framework`

(Note: AWS Lambdas have a similar framework you can install) (https://docs.aws.amazon.com/serverless-application-model/latest/developerguide/serverless-sam-cli-using-debugging.html)

Considerations!

Cold start vs. warm start

- Serverless functions *still* need something to run on, right?
- They essentially run on VMs that aren't necessarily always on



Tips

https://cloud.google.com/functions/docs/bestpractices/tips

(Note - out of all the things that are in flux with cloud apps, this might be the "most" in flux given that it depends on things behind the scenes that are not well-publicized)

(Almost like guessing how the YouTube algorithm works)

(Important) Tips

Idempotent functions

- Functions should **always** return the same output given the same input
  - Even if they use `random`...

HTTP functions must send HTTP response

- If no HTTP response, then a timeout can happen
- Leaving you ... waiting forever (and getting into retry loops)

Dependencies

- Minimize them if possible!
- They need to be installed/loaded **each** time the function cold-starts

More suggestions

Set a *minimum number of instances*

- How many must be "on call" to handle request
- Minimize cold start!

Ignoring the global variable suggestions as those seem - anti-patternish

- E.g., depending on globals to be remembered

Triggers

Different triggers other than HTTPS!

- Pub/Sub, some event
- Here, change in a cloud storage bucket

- Note: must be in same region as bucket

What could we do with this?

Step 1) File uploaded to Cloud Storage

...?

Triggered!

~~(Pub/Sub)~~
~~https://cloud.google.com/run/docs/samples/cloudrun-imageproc-controller~~

~~An offensive image is included above - forewarning it is a zombie scene~~

~~Notes:~~

1) ~~The create bucket code in the documentation is slightly out of date - creating by hand in the Console is less error-prone~~
2) ~~Use python310 for the runtime (I was getting odd syntax errors with 3.12)..b~~
3) ~~Enable Cloud Vision API~~

Lambdas

In CIS437 - F24:

- Activity - AWS Lambda

Worker Pools

New preview thing, but interesting

- Given it is Google - may or may not stick around
- But, headless, potentially-eventless job processing!

(Remember to update the subscription ID before deploying the pool - it is wrong)
(And look at the Console log - the shell log is a bit messy)
https://codelabs.developers.google.com/codelabs/cloud-run/cloud-run-worker-pool-pull-based-subscriptions

In-Class Work

Create a serverless function!

It should:

1) Use your favorite language
2) Have it be HTTP-triggered, unauthenticated, and return HTML
3) When triggered, it must randomly return either a:
    a) Meme picture (make it spooky)
    b) Emoji
    c) ASCII emoji from textfac.es

(i.e., `return random(["meme", "emoji", "ascii-emoji"])`)